

Senior Systems / Process Engineer

Multidisciplinary Engineering — AI Output Validation

EXPERIENCE LEVEL

5 – 15 years

EMPLOYMENT TYPE

Full-time

FUNCTION

Engineering

About the Role

We are building AI agents that generate engineering analysis and drawings across mechanical, electrical, instrumentation, and process disciplines. We need a senior engineer with genuine cross-disciplinary capital projects experience who can independently validate whether an agent's technical output — a P&ID, a load calculation, an electrical schematic, a process flow diagram — is correct. This is not a single-discipline QA role; it requires someone who has worked across multiple engineering disciplines on the same projects and can catch errors that a developer or product manager cannot.

Key Responsibilities

- Review and validate AI-generated engineering outputs across mechanical, electrical, instrumentation, and process domains
- Identify errors, inconsistencies, or non-compliant outputs produced by AI agents
- Cross-check drawings and calculations against relevant codes, standards, and project specifications
- Work directly with AI/product teams to explain, in engineering terms, why an output is incorrect
- Help define validation criteria and error taxonomies as new discipline-specific agents are built
- Interface with clients or stakeholders to clarify technical requirements where needed

Required Qualifications

- Bachelor's degree in Mechanical, Electrical, Instrumentation, Process, or Systems Engineering
- 5–15 years of experience in capital projects (EPC, energy, infrastructure, or process plants)
- Demonstrated cross-disciplinary exposure — not limited to a single discipline silo
- Hands-on experience reading and validating engineering drawings, P&IDs, and design calculations
- Comfortable critiquing technical outputs outside one's primary discipline
- Strong written and verbal communication for explaining technical findings to non-engineers

Preferred Qualifications

- Prior experience as a Systems Engineer, Process Design Engineer, or Lead Engineer spanning multiple disciplines
- Client-facing or stakeholder-facing project experience
- Familiarity with engineering codes and standards relevant to EPC/process industries
- Exposure to digital engineering tools, drawing management systems, or model-based design
- Interest in or exposure to how AI tools are applied in engineering workflows

Soft Skills

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- Strong analytical and troubleshooting ability across technical domains
 - High attention to detail when reviewing technical outputs
 - Comfortable working with cross-functional product and engineering teams
 - Confident, clear communicator able to justify technical judgments